Supplementary Material: Why Early COVID-19 Resists Automated Detection

This document provides extended methodological detail, data integrity results, and training diagnostics to support the main paper. Sections are cross-referenced from the main text where relevant.

1 9-Stage Data Integrity Protocol

We applied the following automated checks to both cohorts before any model training. All checks passed unless otherwise noted.

Check 1 — File Integrity. Verified physical existence and readability of all generated NumPy arrays. Result: 0 missing files (100% integrity across both cohorts).

Check 2 — HU Range Verification. Audited global Hounsfield Unit statistics to confirm the $[-1000, 400]$ HU clipping operated correctly. Result: global mean HU = -614.9 ± 79.1, consistent with air-filled lung parenchyma. Zero extreme values outside the clipping range were detected.

Check 3 — Mask Integrity. Flagged slices with mask area below 1,000 pixels as non-diagnostic. Result: 0 slices failed this criterion after central-quartile slice extraction. Our Middle-30 strategy successfully excluded non-informative apical and basal slices without any additional filtering step.

Check 4 — Slice Sampling Consistency. Audited slices extracted per patient volume. Result: Cohort A mean 21.1 slices/patient (median 22); Cohort B mean 29.3 slices/patient (median 30). The higher yield in Cohort B reflects the more spatially distributed pathology of moderate disease, where more axial slices contain visible parenchymal changes compared to the focal, patchy GGOs of CT-1.

Check 5 — Physics Feature Validation. Compared HU distributions within mask between Normal and COVID cohorts. Result: Normal mean HU = -368.5; Mild COVID mean HU = -338.8 ($\Delta \approx 30$ HU). This shift aligns with the expected GGO density increase. Distributions show significant overlap, as quantified in the main paper (Section 5.3). HU standard deviation: COVID cohort 391.2 vs Normal 405.6, consistent with the hazy homogenisation effect of GGOs blurring the sharp vessel-to-air contrast of healthy parenchyma.

Check 6 — Outlier Detection (IQR Method). Applied Inter-Quartile Range method to all 14 physics features. Result: outlier rates of 3.93% (Mean HU), 3.69% (HU Std Dev), 1.76% (Gradient Mean), and 0.36% (Mask Area). All below 4%, consistent with genuine anatomical variation (patient size, lung inflation) rather than scanner artefact.

Check 7 — Image Quality Audit. Mean gradient magnitude used as sharpness proxy. Dynamic threshold at the 5th percentile flagged potential low-quality scans. Result: 275 slices flagged (5.00%), exactly the expected 5th-percentile rejection rate. No systematic quality bias detected.

Check 8 — Split Balance and Leakage Prevention. Chi-square test applied to label distributions across Train, Validation, and Test splits. Volume-level splitting was enforced: all slices from a patient volume were confined to a single split. Result: χ^2 p-value = 0.521 (no significant label imbalance). No patient volume was found in more than one split.

Check 9 — Severity Gradient Verification. Mann-Whitney U tests on all 14 features for both cohorts. Result: all features show statistically significant differences ($p < 0.0001$) between COVID and Normal distributions. Moderate disease (CT-2) shows larger distributional shifts than mild disease (CT-1) for all features, consistent with the severity gradient hypothesis and the physical expectation that greater GGO burden produces larger whole-lung statistical changes.

2 Full Dataset Statistics

Table 1. MosMedData severity distribution. Heavy skew toward CT-1 reflects real-world clinical epidemiology of COVID-19 presentation.

Class	Description	Cases	%
CT-0	No pneumonia	254	22.8
CT-1	Mild (≤25%)	684	61.6
CT-2	Moderate (25–50%)	125	11.3
CT-3	Severe (50–75%)	45	4.1
CT-4	Critical (≥75%)	2	0.2
Total		**1,110**	**100.0**

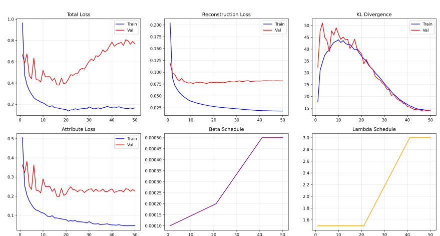
Table 2. Dataset splits and class distribution (slice count). Volume-level splitting ensures no patient anatomy leaks across splits.

	Cohort A		**Cohort B**		**Cohort C**	
Split	**Total**	**COVID**	**Total**	**COVID**	**Total**	**COVID**
Train	3,878	1,928	1,166	550	1,219	605
Validation	804	398	249	147	264	132
Test	818	424	285	153	277	143
Total	**5,500**	**2,750**	**1,700**	**850**	**1,760**	**880**

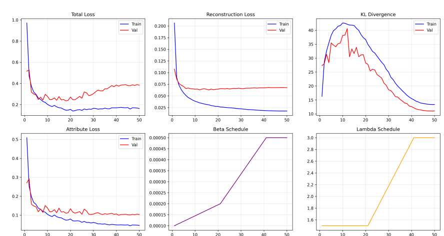
Table 3. Quantitative physics feature comparison by severity level. Signal shift relative to Normal (CT-0) baseline. Moderate disease produces approximately twice the density shift of mild disease. *Gradient Mean shows a non-monotonic severity pattern, precluding a meaningful total shift.

Feature	Normal	Mild	Moderate	Severe
Mean HU (HU)	−368.5	−338.8	−318.4	−305.8
HU Std Dev (HU)	405.7	391.2	370.5	369.3
Gradient Mean	392.0	373.4	361.2	389.7*
Mask Area (pixels)	138,351	148,460	143,405	149,871

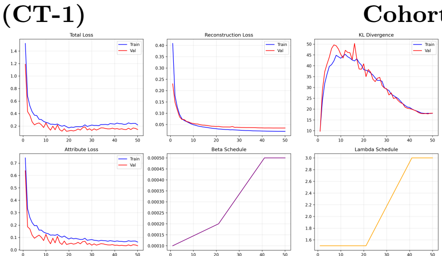
3 Training Dynamics



Cohort A (CT-1)



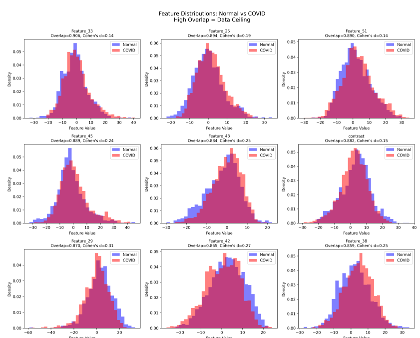
Cohort B (CT-2)



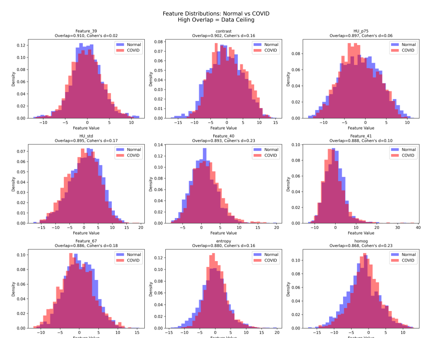
Cohort C (CT-3)

Fig. 1. Training dynamics over 50 epochs. Total loss, reconstruction loss, KL divergence, attribute loss, and annealing schedules (β/λ) are shown. KL stabilizes at ≈17.5 in both cohorts, confirming healthy posterior structure. Attribute loss converges to ≈0.042 on validation. The training-validation gap in Cohort A reflects higher within-split variance from patchy CT-1 GGO distribution.

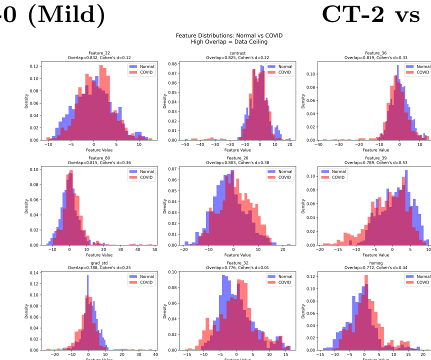
4 Feature Overlap Histograms



CT-1 vs CT-0 (Mild)



CT-2 vs CT-0 (Moderate)



CT-3 vs CT-0 (Severe)

Fig. 2. HU distributions for COVID vs Normal across severity levels. Overlap coefficient (OVL) and Cohen's d are shown per cohort. CT-1: OVL $= 0.845 \pm 0.080$, $d = 0.38 \pm 0.21$. CT-2: OVL $= 0.841 \pm 0.060$, $d = 0.42 \pm 0.15$. CT-3: OVL $= 0.772 \pm 0.050$, $d = 0.368 \pm 0.17$ (learned features). Despite CT-3's higher severity, the learned-feature OVL remains high, confirming a data-intrinsic ceiling rather than a representation failure.

Table 4. Class overlap and separability (mean ± std, 3 seeds). Near-identical values for physics and learned features confirm the ceiling is data-intrinsic, not feature-design-dependent. A Cohen's $d < 0.5$ across 60% of features means no classifier can reliably separate the classes using these features alone. At CT-3, learned feature Cohen's d marginally trails physics features in univariate analysis despite higher classification AUC, consistent with discrimination relying on joint multivariate separation rather than individual feature margins.

Task	Feature Type	Mean OVL	Cohen's d
CT-1 vs CT-0	Physics (14)	0.845 ± 0.080	0.38 ± 0.21
	Learned (top 15)	0.845 ± 0.070	0.41 ± 0.19
CT-2 vs CT-0	Physics (14)	0.841 ± 0.060	0.42 ± 0.15
	Learned (top 15)	0.841 ± 0.090	0.44 ± 0.18
CT-3 vs CT-0	Physics (14)	0.776 ± 0.050	0.38 ± 0.15
	Learned (top 15)	0.783 ± 0.014	0.368 ± 0.17

Across severity levels, per-feature overlap decreases progressively from CT-1 to CT-3 and Cohen's d increases correspondingly, yet the magnitude of this univariate shift does not proportionally account for the AUC jump from 67.4% to 99.3%. Near-perfect CT-3 detection relies on joint separation across the full physics-constrained latent space — a multivariate signal that emerges when GGO burden exceeds 50% but remains invisible to single-feature analysis

5 Full PAR-VAE Classification Results

Table 5. Full PAR-VAE and CNN classification results across all severity levels (mean ± std over 3 seeds). Bold marks best test AUC per task.

Task	Model	Val Acc	Test Acc	Test F1	Test AUC
CT-1 vs CT-0	PAR-VAE (LogReg)	62.0 ± 2.1	62.6 ± 2.8	65.8 ± 2.6	67.4 ± 1.4
(Mild)	CNN Baseline	66.8 ± 1.5	65.2 ± 0.9	68.2 ± 2.5	$\mathbf{69.8 \pm 1.1}$
CT-2 vs CT-0	PAR-VAE (LogReg)	70.8 ± 1.2	66.5 ± 1.6	69.3 ± 2.0	$\mathbf{74.6 \pm 0.8}$
(Moderate)	CNN Baseline	68.3 ± 2.4	64.1 ± 2.1	66.7 ± 1.9	70.0 ± 1.5
CT-3 vs CT-0	PAR-VAE (RBF-SVM)	98.5 ± 1.1	97.3 ± 2.9	96.70 ± 3.79	$\mathbf{99.3 \pm 1.0}$
(Severe)	CNN Baseline	57.1 ± 6.7	60.6 ± 6.3	57.22 ± 11.58	66.0 ± 7.5

6 Physics Feature Logistic Regression Coefficients

Tables 6–8 report mean ± std logistic regression coefficients across seeds 16, 42, and 999 for the 14-dimensional physics feature classifier on each cohort. Features are sorted by mean absolute coefficient magnitude. The high seed-to-seed variance (large std relative to mean) is itself diagnostic: no single feature has stable discriminative power, and the classifier redistributes weight across correlated features depending on initialisation. This instability is a direct consequence of the near-unity overlap reported in Table 4. For CT-3, high variance with near-perfect accuracy reflects solution non-uniqueness in a well-separated regime — any of the three seed solutions achieves $> 95\%$ test accuracy, confirming the variance is geometric rather than a symptom of overfitting.

Table 6. Physics feature logistic regression coefficients — CT-1 vs CT-0 (Mild). Mean $\pm$ std across 3 seeds, sorted by mean |coeff|.

Feature	**Mean Coeff $\pm$ Std**	**Mean |Coeff|**
`Gradient Std. Dev`	-0.517 ± 0.768	0.803
`Texture Entropy`	-0.086 ± 0.764	0.734
`HU Std. Dev`	$+0.108 \pm 0.744$	0.714
`Homogeneity`	$+0.651 \pm 0.459$	0.651
`Fractional Occupancy`	$+0.283 \pm 0.656$	0.523
`HU 25th Percentile`	$+0.074 \pm 0.552$	0.461
`Gradient Mean`	-0.454 ± 0.205	0.454
`HU 50th Percentile`	$+0.217 \pm 0.526$	0.423
`Mean HU`	$+0.384 \pm 0.063$	0.384
`HU 75th Percentile`	$+0.331 \pm 0.270$	0.342
`HU 90th Percentile`	-0.326 ± 0.256	0.339
`Mask Area`	$+0.143 \pm 0.305$	0.335
`HU 10th Percentile`	$+0.040 \pm 0.334$	0.280
`GLCM Contrast`	$+0.081 \pm 0.197$	0.206

Table 7. Physics feature logistic regression coefficients — CT-2 vs CT-0 (Moderate). Mean ± std across 3 seeds, sorted by mean |coeff|. `HU 90th Percentile` has the largest mean magnitude but also the largest variance (±1.114), indicating highly unstable weighting across seeds.

Feature	Mean Coeff ± Std	Mean |Coeff|
`HU 90th Percentile`	-0.251 ± 1.114	1.133
`Fractional Occupancy`	$+0.701 \pm 0.279$	0.701
`Mean HU`	-0.309 ± 0.692	0.666
`Gradient Std. Dev`	$+0.046 \pm 0.599$	0.545
`HU 50th Percentile`	$+0.336 \pm 0.558$	0.544
`HU Std. Dev`	-0.383 ± 0.630	0.530
`HU 25th Percentile`	$+0.066 \pm 0.445$	0.434
`GLCM Contrast`	$+0.147 \pm 0.454$	0.406
`Mask Area`	-0.142 ± 0.392	0.401
`Texture Entropy`	$+0.396 \pm 0.237$	0.396
`Gradient Mean`	$+0.146 \pm 0.451$	0.367
`HU 75th Percentile`	$+0.238 \pm 0.355$	0.340
`HU 10th Percentile`	-0.305 ± 0.231	0.319
`Homogeneity`	-0.009 ± 0.227	0.210

Table 8. Physics feature logistic regression coefficients — CT-3 vs CT-0 (Severe). Mean ± std across 3 seeds, sorted by mean |coeff|. Despite high coefficient variance, all three seeds achieve > 95% test accuracy, confirming variance reflects geometric non-uniqueness of the decision boundary in a well-separated regime rather than estimator instability.

Feature	Mean Coeff ± Std	Mean |Coeff|
`Fractional Occupancy`	$+0.401 \pm 1.387$	1.423
`Mask Area`	-0.049 ± 1.410	1.257
`Texture Entropy`	-0.042 ± 1.417	1.189
`HU 75th Percentile`	$+0.684 \pm 0.901$	1.067
`HU 10th Percentile`	-0.527 ± 0.891	0.978
`HU 90th Percentile`	-0.001 ± 0.886	0.835
`HU 50th Percentile`	$+0.827 \pm 0.492$	0.827
`Homogeneity`	-0.104 ± 0.797	0.689
`Gradient Std. Dev`	$+0.685 \pm 0.917$	0.685
`Gradient Mean`	-0.235 ± 0.574	0.615
`HU 25th Percentile`	$+0.067 \pm 0.610$	0.519
`Mean HU`	-0.476 ± 0.311	0.476
`HU Std. Dev`	$+0.114 \pm 0.492$	0.426
`GLCM Contrast`	-0.103 ± 0.237	0.189